

G4 Differentiation rules A level only

Name: _____

Class: _____

Date: _____

Time: **386 minutes**

Marks: **322 marks**

Comments:

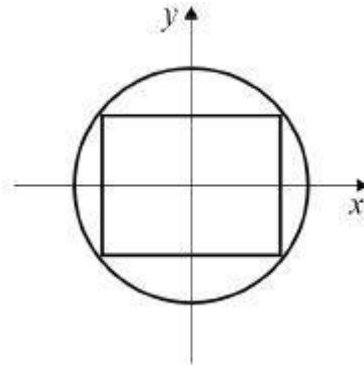
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Q1.

A company is designing a logo. The logo is a circle of radius 4 inches with an inscribed rectangle. The rectangle must be as large as possible.

The company models the logo on an x - y plane as shown in the diagram.



Use calculus to find the maximum area of the rectangle.

Fully justify your answer.

(Total 10 marks)

Q2.

Find the coordinates of the stationary point of the curve with equation

$$(x + y - 2)^2 = e^y - 1$$

(Total 7 marks)

Q3.

A function f is defined by $f(x) = \frac{x}{\sqrt{2x-2}}$

(a) State the maximum possible domain of f .

(2)

(b) Use the quotient rule to show that $f'(x) = \frac{x-2}{(2x-2)^{\frac{3}{2}}}$

(3)

(c) Show that the graph of $y = f(x)$ has exactly one point of inflection.

(7)

(d) Write down the values of x for which the graph of $y = f(x)$ is convex.

(1)

(Total 13 marks)

Q4.

A curve has equation $y = \frac{2x+3}{4x^2+7}$

- (a) (i) Find $\frac{dy}{dx}$ (2)
- (ii) Hence show that y is increasing when $4x^2 + 12x - 7 < 0$ (4)
- (b) Find the values of x for which y is increasing. (2)

(Total 8 marks)

Q5.

A curve is defined by the parametric equations

$$x = t^3 + 2, \quad y = t^2 - 1$$

- (a) Find the gradient of the curve at the point where $t = -2$ (4)
- (b) Find a Cartesian equation of the curve. (2)

(Total 6 marks)

Q6.

A curve has equation $y = 2x \cos 3x + (3x^2 - 4) \sin 3x$

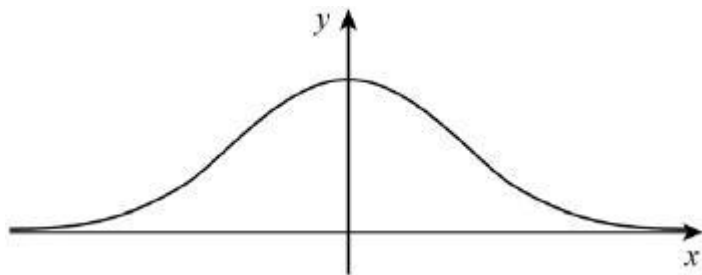
- (a) Find $\frac{dy}{dx}$, giving your answer in the form $(mx^2 + n) \cos 3x$, where m and n are integers. (4)
- (b) Show that the x -coordinates of the points of inflection of the curve satisfy the equation

$$\cot 3x = \frac{9x^2 - 10}{6x}$$

(4)
(Total 8 marks)

Q7.

The diagram shows part of the graph of $y = e^{-x^2}$

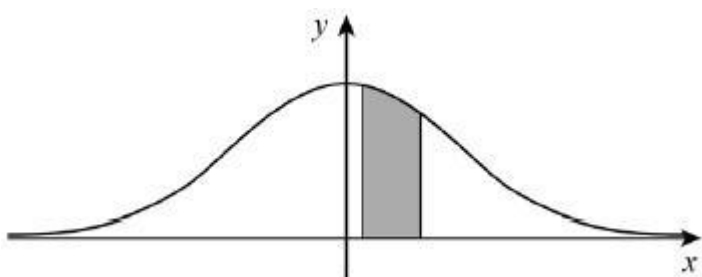


The graph is formed from two convex sections, where the gradient is increasing, and one concave section, where the gradient is decreasing.

- (a) Find the values of x for which the graph is concave.

(4)

- (b) The finite region bounded by the x -axis and the lines $x = 0.1$ and $x = 0.5$ is shaded.



Use the trapezium rule, with 4 strips, to find an estimate for $\int_{0.1}^{0.5} e^{-x^2} dx$

Give your estimate to four decimal places.

(3)

- (c) Explain with reference to your answer in part (a), why the answer you found in part (b) is an underestimate.

(2)

- (d) By considering the area of a rectangle, and using your answer to part (b), prove that the shaded area is 0.4 correct to 1 decimal place.

(3)

(Total 12 marks)

Q8.

- (a) Find $\frac{dy}{dx}$ when

$$y = e^{3x} + \ln x$$

(2)

- (b) (i) Given that $u = \frac{\sin x}{1 + \cos x}$, show that $\frac{du}{dx} = \frac{1}{1 + \cos x}$.

(3)

- (ii) Hence show that if $y = \ln \left(\frac{\sin x}{1 + \cos x} \right)$, then $\frac{dy}{dx} = \operatorname{cosec} x$.

(2)

(Total 7 marks)

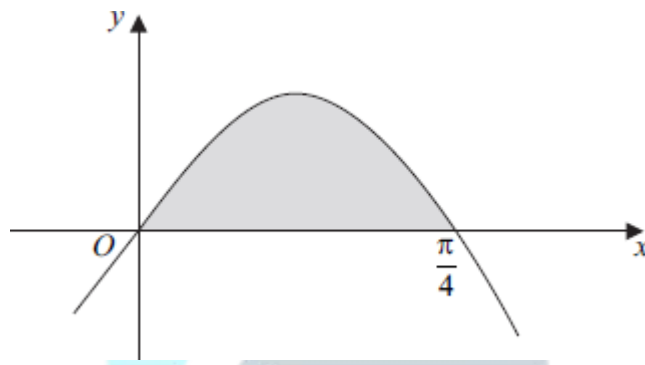
Q9.

A curve has equation $y = 4x \cos 2x$.

- (a) Find an exact equation of the tangent to the curve at the point on the curve where $x = \frac{\pi}{4}$.

(5)

- (b) The region shaded on the diagram below is bounded by the curve $y = 4x \cos 2x$ and the x -axis from $x = 0$ to $x = \frac{\pi}{4}$.



By using integration by parts, find the exact value of the area of the shaded region.

(5)

(Total 10 marks)

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Q10.

- (a) Given that $y = x^4 \tan 2x$, find $\frac{dy}{dx}$.

(3)

- (b) Find the gradient of the curve with equation $y = \frac{x^2}{x-1}$ at the point where $x = 3$.

(3)

(Total 6 marks)

Q11.

- (a) Given that $x = \frac{1}{\sin \theta}$, use the quotient rule to show that $\frac{dx}{d\theta} = -\operatorname{cosec} \theta \cot \theta$.

(3)

- (b) Use the substitution $x = \operatorname{cosec} \theta$ to find $\int_{\sqrt{2}}^2 \frac{1}{x^2 \sqrt{x^2 - 1}} dx$, giving your answer to three significant figures.

(9)

(Total 12 marks)

Q12.

A curve has equation $y = x^3 \ln x$.

- (a) Find $\frac{dy}{dx}$.
- (b) (i) Find an equation of the tangent to the curve $y = x^3 \ln x$ at the point on the curve where $x = e$.
- (ii) This tangent intersects the x -axis at the point A . Find the exact value of the x -coordinate of the point A .

(2)

(3)

(2)

(Total 7 marks)

Q13.

- (a) Given that $x = \frac{\sin y}{\cos y}$, use the quotient rule to show that

$$\frac{dx}{dy} = \sec^2 y$$

(3)

- (b) Given that $\tan y = x - 1$, use a trigonometrical identity to show that

$$\sec^2 y = x^2 - 2x + 2$$

(2)

- (c) Show that, if $y = \tan^{-1}(x - 1)$, then

$$\frac{dy}{dx} = \frac{1}{x^2 - 2x + 2}$$

(1)

- (d) A curve has equation $y = \tan^{-1}(x - 1) - \ln x$.

- (i) Find the value of the x -coordinate of each of the stationary points of the curve.

(4)

(ii) Find $\frac{d^2y}{dx^2}$.

(2)

(iii) Hence show that the curve has a minimum point which lies on the x -axis.

(2)

(Total 14 marks)

Q14.

(a) Find $\frac{dy}{dx}$ when $y = (x^3 - 1)^6$.

(2)

(b) A curve has equation $y = x \ln x$.

(i) Find $\frac{dy}{dx}$.

(2)

(ii) Find an equation of the tangent to the curve $y = x \ln x$ at the point on the curve where $x = e$.

(3)

(Total 7 marks)

Q15.

(a) Given that $x = \tan(3y + 1)$:

(i) find $\frac{dx}{dy}$ in terms of y ;

(2)

(ii) find the value of $\frac{dy}{dx}$ when $y = -\frac{1}{3}$.

(2)

(b) Sketch the graph of $y = \tan^{-1} x$.

(2)

(Total 6 marks)

Q16.

(a) (i) Find $\frac{dy}{dx}$ when $y = xe^{2x}$.

(3)

(ii) Find an equation of the tangent to the curve $y = xe^{2x}$ at the point $(1, e^2)$

(2)

- (b) Given that $y = \frac{2 \sin 3x}{1 + \cos 3x}$, use the quotient rule to show that

$$\frac{dy}{dx} = \frac{k}{1 + \cos 3x}$$

where k is an integer.

(4)

(Total 9 marks)

Q17.

A curve is defined by the equation $2y + e^{2y^2} = x^2 + C$, where C is a constant.

The point $P = \left(1, \frac{1}{e}\right)$ lies on the curve.

- (a) Find the exact value of C .

(1)

- (b) Find an expression for $\frac{dy}{dx}$ in terms of x and y .

(7)

- (c) Verify that $P = \left(1, \frac{1}{e}\right)$ is a stationary point on the curve.

(2)

(Total 10 marks)

Q18.

A curve has equation $y = e^{-4x}(x^2 + 2x - 2)$.

- (a) Show that $\frac{dy}{dx} = 2e^{-4x}(5 - 3x - 2x^2)$.

(3)

- (b) Find the exact values of the coordinates of the stationary points of the curve.

(5)

(Total 8 marks)

Q19.

It is given that $y = \tan 4x$.

- (a) By writing $\tan 4x$ as $\frac{\sin 4x}{\cos 4x}$, use the quotient rule to show that $\frac{dy}{dx} = p(1 + \tan^2 4x)$, where p is a number to be determined.

(3)

- (b) Show that $\frac{d^2y}{dx^2} = q(1 + y^2)$, where q is a number to be determined.

(5)

(Total 8 marks)

Q20.

A curve is defined by the equation

$$x^2 + xy = e^y$$

Find the gradient at the point $(-1, 0)$ on this curve.

(Total 5 marks)

Q21.

- (a) (i) Express $\sin 2\theta$ and $\cos 2\theta$ in terms of $\sin \theta$ and $\cos \theta$.

(2)

- (ii) Given that $0 < \theta < \frac{\pi}{2}$ and $\cos \theta = \frac{3}{5}$, show that $\sin 2\theta = \frac{24}{25}$ and find the value of $\cos 2\theta$.

(2)

- (b) A curve has parametric equations

$$x = 3 \sin 2\theta, \quad y = 4 \cos 2\theta$$

- (i) Find $\frac{dy}{dx}$ in terms of θ .

(3)

- (ii) At the point P on the curve, $\cos \theta = \frac{3}{5}$ and $0 < \theta < \frac{\pi}{2}$. Find an equation of the tangent to the curve at the point P .

(3)

(Total 10 marks)

Q22.

(a) Find $\frac{dy}{dx}$ when:

(i) $y = \ln(5x - 2)$;

(2)

(ii) $y = \sin 2x$.

(2)

(b) The functions f and g are defined with their respective domains by

$$f(x) = \ln(5x - 2), \quad \text{for real values of } x \text{ such that } x \geq \frac{1}{2}$$

$$g(x) = \sin 2x, \quad \text{for real values of } x \text{ in the interval } -\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$$

(i) Find the range of f .

(2)

(ii) Find an expression for $gf(x)$.

(1)

(iii) Solve the equation $gf(x) = 0$.

(3)

(iv) The inverse of g is g^{-1} . Find $g^{-1}(x)$.

(2)

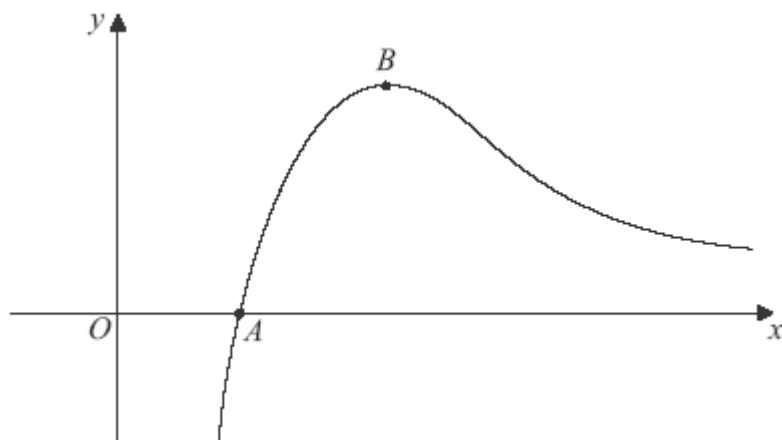
(Total 12 marks)

Q23.

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The diagram shows the curve $y = \frac{\ln x}{x}$.



The curve crosses the x -axis at A and has a stationary point at B .

- (a) State the coordinates of A . (1)
- (b) Find the coordinates of the stationary point, B , of the curve, giving your answer in an exact form. (5)
- (c) Find the exact value of the gradient of the normal to the curve at the point where $x = e^3$. (3)
- (Total 9 marks)**

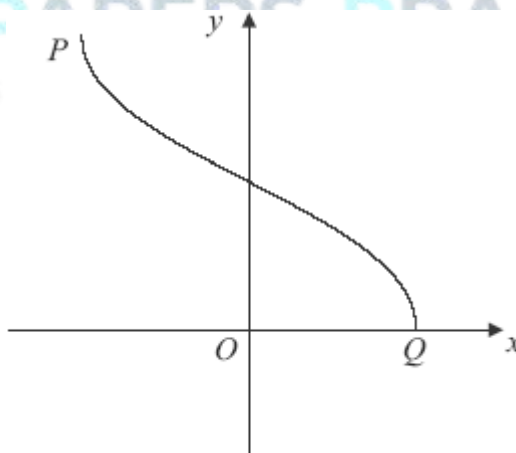
Q24.

A curve has equation $y = e^{2x}(x^2 - 4x - 2)$.

- (a) Find the value of the x -coordinate of each of the stationary points of the curve. (6)
- (b) (i) Find $\frac{d^2y}{dx^2}$. (2)
- (ii) Determine the nature of each of the stationary points of the curve. (2)
- (Total 10 marks)**

Q25.

The sketch shows the graph of $y = \cos^{-1} x$.



- (a) Write down the coordinates of P and Q , the end points of the graph. (2)
- (b) Describe a sequence of two geometrical transformations that maps the graph of $y = \cos^{-1} x$ onto the graph of $y = 2 \cos^{-1}(x - 1)$.

(4)

- (c) Sketch the graph of $y = 2 \cos^{-1}(x - 1)$.

(2)

- (d) (i) Write the equation $y = 2 \cos^{-1}(x - 1)$ in the form $x = f(y)$.

(2)

- (ii) Hence find the value of $\frac{dx}{dy}$ when $y = 2$.

(3)

(Total 13 marks)

Q26.

- (a) Given that $y = \frac{4x}{4x-3}$, use the quotient rule to show that $\frac{dy}{dx} = \frac{k}{(4x-3)^2}$, where k is an integer.

(2)

- (b) (i) Given that $y = x \ln(4x - 3)$, find $\frac{dy}{dx}$.

(3)

- (ii) Find an equation of the tangent to the curve $y = x \ln(4x - 3)$ at the point where $x = 1$.

(3)

- (c) (i) Use the substitution $u = 4x - 3$ to find $\int \frac{4x}{4x-3} dx$, giving your answer in terms of x .

(4)

- (ii) By using integration by parts, or otherwise, find $\int \ln(4x - 3) dx$.

(4)

(Total 16 marks)

Q27.

- (a) The curve with equation

$$y = \frac{\cos x}{2x+1}, \quad x > -\frac{1}{2}$$

intersects the line $y = \frac{1}{2}$ at the point where $x = \alpha$.

- (i) Show that α lies between 0 and $\frac{\pi}{2}$. (2)

- (ii) Show that the equation $\frac{\cos x}{2x+1} = \frac{1}{2}$ can be rearranged into the form

$$x = \cos x - \frac{1}{2}$$

(1)

- (iii) Use the iteration $x_{n+1} = \cos x_n - \frac{1}{2}$ with $x_1 = 0$ to find x_3 , giving your answer to three decimal places. (2)

- (b) (i) Given that $y = \frac{\cos x}{2x+1}$, use the quotient rule to find an expression for $\frac{dy}{dx}$. (3)

- (ii) Hence find the gradient of the normal to the curve $y = \frac{\cos x}{2x+1}$ at the point on the curve where $x = 0$. (2)
- (Total 10 marks)

Q28.

A curve is defined by the equation $4x^2 + y^2 = 4 + 3xy$.

Find the gradient at the point (1, 3) on this curve.

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(Total 5 marks)

Q29.

- (a) Find $\frac{dy}{dx}$ when:

(i) $y = (2x^2 - 5x + 1)^{20}$; (2)

(ii) $y = x \cos x$. (2)

- (b) Given that

$$y = \frac{x^3}{x-2}$$

show that

$$\frac{dy}{dx} = \frac{kx^2(x-3)}{(x-2)^2}$$

where k is a positive integer.

(3)

(Total 7 marks)

Q30.

A curve has equation $3xy - 2y^2 = 4$.

Find the gradient of the curve at the point (2, 1).

(Total 5 marks)

Q31.

Find $\frac{dy}{dx}$ when:

(a) $y = (3x + 1)^5$;

(2)

(b) $y = \ln(3x + 1)$;

(2)

(c) $y = (3x + 1)^5 \ln(3x + 1)$.

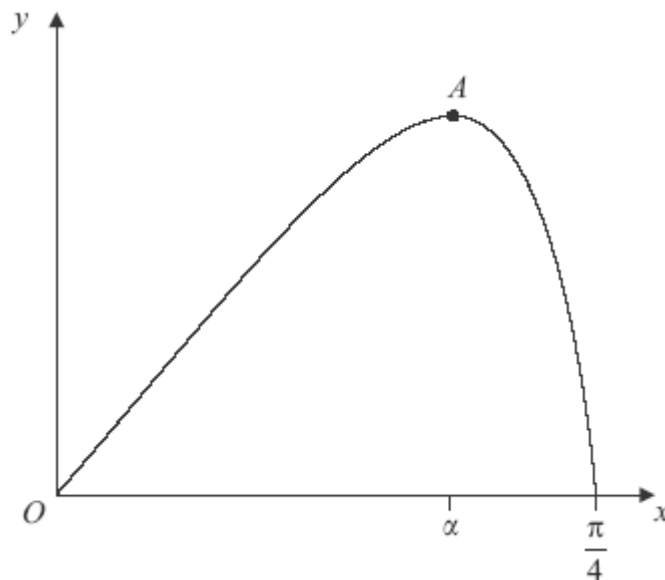
(3)

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(Total 7 marks)

Q32.

A curve is defined for $0 \leq x \leq \frac{\pi}{4}$ by the equation $y = x \cos 2x$, and is sketched below.



- (a) Find $\frac{dy}{dx}$. (2)
- (b) The point A , where $x = \alpha$, on the curve is a stationary point.
- (i) Show that $1 - 2\alpha \tan 2\alpha = 0$. (2)
- (ii) Show that $0.4 < \alpha < 0.5$. (2)
- (iii) Show that the equation $1 - 2x \tan 2x = 0$ can be rearranged to become $x = \frac{1}{2} \tan^{-1}\left(\frac{1}{2x}\right)$. (1)
- (iv) Use the iteration $x_{n+1} = \frac{1}{2} \tan^{-1}\left(\frac{1}{2x_n}\right)$ with $x_1 = 0.4$ to find x_3 , giving your answer to two significant figures. (2)
- (c) Use integration by parts to find $\int_0^{0.5} x \cos 2x dx$, giving your answer to three significant figures. (5)

(Total 14 marks)

Q33.

- (a) Given that $y = \frac{\sin \theta}{\cos \theta}$, use the quotient rule to show that $\frac{dy}{d\theta} = \sec^2 \theta$. (3)
- (b) Given that $x = \sin \theta$, show that $\frac{x}{\sqrt{1-x^2}} = \tan \theta$. (2)
- (c) Use the substitution $x = \sin \theta$ to find $\int \frac{1}{(1-x^2)^{\frac{3}{2}}} dx$, giving your answer in terms of x . (5)
- (Total 10 marks)

Q34.

- (a) Find $\frac{dy}{dx}$ when:
- (i) $y = (4x^2 + 3x + 2)^{10}$; (2)
- (ii) $y = x^2 \tan x$. (2)
- (b) (i) Find $\frac{dx}{dy}$ when $x = 2y^3 + \ln y$. (1)
- (ii) Hence find an equation of the tangent to the curve $x = 2y^3 + \ln y$ at the point (2,1). (3)
- (Total 8 marks)

Q35.

- (a) (i) Solve the differential equation $\frac{dy}{dt} = y \sin t$ to obtain y in terms of t . (4)
- (ii) Given that $y = 50$ when $t = \pi$, show that $y = 50e^{-(1 + \cos t)}$. (3)
- (b) A wave machine at a leisure pool produces waves. The height of the water, y cm, above a fixed point at time t seconds is given by the differential equation

$$\frac{dy}{dt} = y \sin t$$

- (i) Given that this height is 50 cm after π seconds, find, to the nearest centimetre, the height of the water after 6 seconds.

(2)

- (ii) Find $\frac{d^2 y}{dt^2}$ and hence verify that the water reaches a maximum height after π seconds.

(4)

(Total 13 marks)



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